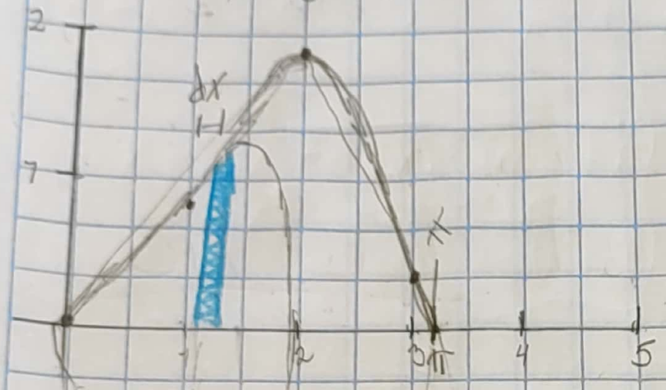
Hoja de Trabajo 3

Ejercicio 1

$y = x\,\text{sen}\,x,\quad y=0,\ x=0,\ x=\pi$



$m = 2x$
$dm = 2\,dx$

$\cos m\ \frac{dm}{2}$

$\cos m\ \frac{dm}{2}$

$V = \pi r^2\,dx$
$V = \pi\,(x\,\text{sen}\,x)^2\,dx$
$V = \pi x^2\,\text{sen}^2 x\,dx$

$V = \pi\int_0^{\pi} x^2\,\text{sen}^2 x\,dx$

$V = \pi\int_0^{\pi} x^2\left(\frac{1-\cos 2x}{2}\right)dx$

$V = \pi\int_0^{\pi} x^2 - x^2\cos 2x\,dx = \frac{\pi}{2}\left[\int x^2\,dx - \int x^2\cos 2x\,dx\right]$

$U = x^2 \qquad dV = \cos 2x\,dx$
$dU = 2x\,dx \qquad V = \frac{1}{2}\,\text{sen}\,2x$

$-\left[x^2\cdot\frac{1}{2}\,\text{sen}\,2x - \int\frac{1}{2}\,\text{sen}\,2x\cdot 2x\,dx\right]$

$-\left[\frac{x^2}{2}\,\text{sen}\,2x - \int \text{sen}\,2x\cdot x\,dx\right]$

$U' = x \qquad dV = \text{sen}\,2x$
$dU = dx \qquad V = \frac{1}{2}\cos 2x$

$-\left(x+\frac{1}{2}\cos 2x - \int -\frac{1}{2}\cos 2x\,dx\right.$

$+\frac{x}{2}\cos 2x + \frac{1}{2}\int\cos 2x\,dx$

$m = 2x$
$dm = 2\,dx$

$-\left[\frac{x^2}{2}\,\text{sen}\,2x + \frac{x}{2}\cos 2x - \frac{1}{4}\,\text{sen}\,2x\right]$

$\frac{\pi}{2}\left[\frac{x^3}{3} - \frac{x^2}{2}\,\text{sen}\,2x - \frac{x}{2}\cos 2x + \frac{1}{4}\,\text{sen}\,2x\right]\Big|_0^{\pi}$

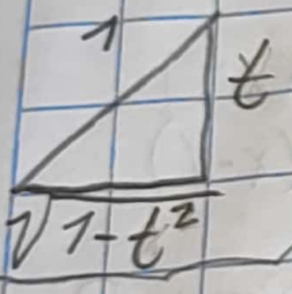
$V = 13.7674\,u^3$

Ejercicio 2

$$\int_0^{\frac{\sqrt{3}}{2}} \frac{t^2}{(1-t^2)^{3/2}}\,dt = \int \frac{\text{sen}^2\theta}{(1-\text{sen}^2\theta)^{3/2}}\cos\theta = \int \frac{\text{sen}^2\theta\cos\theta}{(\cos^2\theta)^{3/2}}\,d\theta$$

$t = 1\,\text{sen}\,\theta \qquad dt = \cos\theta$

$\frac{t}{1} = \text{sen}\,\theta \qquad \text{sen}^{-1} t = \theta$



$$\int \frac{\text{sen}^2\theta\cos\theta}{\cos^3\theta}\,d\theta = \int \frac{\text{sen}^2\theta}{\cos^2\theta}\,d\theta = \int \tan^2\theta\,d\theta$$

$$= \int \sec^2\theta - 1\,d\theta = \int \sec^2\theta\,d\theta - \int 1\,d\theta$$

$$= \tan\theta - \theta$$

$$= \tan\theta - \text{sen}^{-1} t$$

$$= \frac{t}{\sqrt{1-t^2}} - \text{sen}^{-1} t \Big|_0^{\frac{\sqrt{3}}{2}} \approx 0.6848$$

Ejercicio 3

$$\int \frac{x^5-10x^3}{x^4-10x^2+9}dx = \int x - \frac{9x+9}{x^4-10x^2+9} = \int x\,dx - \int \frac{9x}{x^4-10x^2+9}$$

$$\int x\,dx - \int \frac{A}{x-3}dx + \int \frac{B}{x+3}dx + \int \frac{C}{x-1}dx + \int \frac{D}{x+1}dx$$

$$\begin{array}{r|l} & x \\ \hline x^4-10x^2+9 & x^5+0-10x^3+0+0-10 \\ & -x^5 \quad +10x^3-9x \\ \hline & \quad\quad\quad -9x \end{array}$$

$D(x-3)(x+3)(x-1)$

$x^4-9x^2-x^2+9$

$x^2(x^4-x^2)(-9x^2+9)$

$x^2(x^2-1)-9(x^2-1)$

$(x^2-9)(x^2-1)$

$(x-3)(x+3)(x-1)(x+1)$

$$\frac{A(x+3)(x-1)(x+1)+B(x+3)(x-1)(x+1)+C(x-3)(x+3)(x+1)+D(x-3)}{(x-3)(x+3)(x-1)(x+1)}$$

$A(x+3)(x^2-1)+B(x-3)(x^2-1)+C(x^2-9)(x+1)+D(x^2-9)(x-1)$

$\cancel{Ax^3}-\cancel{Ax}+\cancel{3Ax^2}-\cancel{3A}+\cancel{Bx^3}-\cancel{Bx}-\cancel{3Bx^2}+\cancel{3B}+\cancel{Cx^3}+\cancel{Cx^2}-\cancel{9Cx}-\cancel{9C}+\cancel{Dx^3}-\cancel{Dx^2}-\cancel{9Dx}+\cancel{9D}$

$A+B+C+D=0$

$3A-3B+C-D=0$

$-A-9C-9D-B=9$

$-3A+3B-9D-9C=0$

$A=-9/16$

$B=-9/16$

$C=9/16$

$D=9/16$

$A+B+C+D=0$

$3A-3B+C-D=0$

$-A-B-9C-9D=9$

$-3A+3B-9C+9D=0$

$$\begin{array}{r} x-3 \\ x+3 \\ \hline x^2-3x \\ +3x-9 \\ \hline x^2-9 \\ x-1 \\ \hline x^3-9x-x^2+9 \end{array}$$

$$\int x\,dx + \frac{9}{16}\int \frac{1}{x-3}\,dx - \frac{9}{16}\int \frac{1}{x+3}\,dx + \frac{9}{16}\int \frac{1}{x-1}\,dx + \frac{9}{16}\int \frac{1}{x+1}\,dx$$

$$\frac{x^2}{2} - \ln|x-3| - \frac{9}{16}\ln|x+3| + \frac{9}{16}\ln|x-1| + \frac{9}{16}\ln|x+1| + C$$

Ejercicio 4

$$\int_1^{\infty} \sqrt{x}\, e^{-\sqrt{x}}\,dx = \int \sqrt{x}\, e^{m} \frac{dm}{-\frac{1}{2\sqrt{x}}} = \int \sqrt{x}\, e^{m}(2m)\,dm = \int -\sqrt{x}\, e^{m}(2m)\,dm$$

$m = -\sqrt{x}$

$dm = -\frac{1}{2\sqrt{x}}dx$

$$= \int m e^{m}(-2m)\,dm = \int 2m^2 e^{m}\,dm = -2\int m^2 e^{m}\,dm$$

$U = m^2 \qquad dv = e^{m}$

$dU = 2m\,dm \qquad V = e^{m}$

$$-2\left[m^2 e^{m} - \int e^{m} 2m\,dm\right]$$

$$-2\int e^{m} m\,dm$$

$$-2\left(me^{m} - \int e^{m}\,dm\right)$$

$$-2me^{m} - e^{-\sqrt{x}}$$

$U = m \qquad dV = e^{m}$

$dU = dm \qquad V = e^{m}$

$$-2\left[(-\sqrt{x})^2 e^{-\sqrt{x}} - 2(-\sqrt{x})e^{-\sqrt{x}} - e^{-\sqrt{x}}\right]$$

$$-2\left[-xe^{-\sqrt{x}} - 2(-\sqrt{x}\,e^{-\sqrt{x}} - e^{-\sqrt{x}})\right]\Big|_1^{\infty}$$

$$-2\left[-1e^{-\sqrt{1}} - 2(-\sqrt{1}\,e^{-\sqrt{1}} - e^{-\sqrt{1}})\right] - [$$

$$0 - (-3.6787) = \underline{3.6787}$$